

QUIZ #2

*NOTE: Reference equations on the last page.
Good luck!*

1. Health Physics – Matching:Column A

- _____ 1. Roentgen (R)
- _____ 2. RAD
- _____ 3. REM
- _____ 4. Curie (Ci)
- _____ 5. Becquerel (Bq)
- _____ 6. Geiger Counter
- _____ 7. Ion Chamber
- _____ 8. REM Ball

Column B

- A. Absorbed dose in energy (ergs or MeV) per gram **of tissue.**
- B. Sensitive to $\beta/\gamma/x$ and utilized primarily for area radiation surveys.
- C. Absorbed dose in energy (ergs or MeV) per gram of matter.
- D. Amount of radioactivity in 1 gram of Radium.
- E. 1 decay per second (DPS).
- F. Sensitive to $\alpha/\beta/\gamma/x$ and utilized primarily for contamination surveys.
- G. Exposure in coulombs/cc of air.
- H. Sensitive to fast neutrons; reads out in mREM/hr.

2. You have a 1 Ci source of Cesium-137. The primary gamma decay energy is 661.7 keV, with a likelihood of 94.7%. What is the exposure rate (R/hr) at a distance of 1 foot? Using the previously calculated dose rate and the **inverse square law**, what would the exposure rate be at 5 feet from the same source? Show your work.

3. During a reactor startup, the initial NFM count-rate $CR_0 = 15$ CPS at time zero ($t=0$) and the initial value for $\rho_{\text{net}} = -8100$ pcm with the control rods seated (reactor fully shutdown).
- a) Fill in the values for K_{eff} and reactivity ρ in the following table as rods are withdrawn and the NFM count-rate is approximately doubled 5X – ***SHOW ALL WORK! Calculate k_{eff} and ρ values to 5 decimal places (e.g. 0.00001)***:

Time	NFM (CPS)	K_{eff}	Reactivity ρ_{net} ($\Delta k/k$)
0	$CR_0 = 15$		
1	$CR_1 = 35$		
2	$CR_2 = 80$		
3	$CR_3 = 180$		
4	$CR_4 = 600$		
5	$CR_5 = 4000$		

- b) At timestep #5 (after the 5th double), is the reactor super-critical or sub-critical? By how much net reactivity ρ_{net} (pcm)?

4. Answer TRUE or FALSE to the following:

- _____ a. The prompt neutron generation time (ℓ_p) is less than the delayed neutron generation time (ℓ_d).
- _____ b. The prompt neutron fraction ($1-\beta$) is less than the delayed neutron fraction (β).
- _____ c. Equal changes in K_{eff} result in equal changes in the equilibrium neutron population N_{eq} for a subcritical reactor.
- _____ d. As K_{eff} approaches 1.0, more generations are required to reach equilibrium; therefore, less time is required to reach equilibrium.
- _____ e. If K_{eff} is increased halfway to 1.0 from its initial value, the neutron count rate will increase by half.
- _____ f. Reactor period τ is the time it takes for reactor power to change by a factor of 3.14 (or 'e').
- _____ g. RF, Microwave, and Infrared are examples of ionizing radiation.
- _____ h. In Stochastic health effects of radiation, the severity increases with dose.
- _____ i. Minimizing the time spent near a radiation source would be consistent with As Low As Reasonably Achievable (ALARA) principles.

5. If $\ell_p = 3 \times 10^{-5}$ sec, $\ell_d = 12.7$ sec, $\beta = 0.0073$, and 100 pcm of reactivity is added to the core from an initially critical state (i.e. $\rho_{net} = +100$ pcm), calculate the final power in Watts (P_f) at $t = 60$ sec if $P_i = 250$ Watts.

See lab questions on the next page

HP Lab Questions:

4. What are the limits for a Radiation Area and a High Radiation Area in mrem/hr?
5. What are two important pre-operational checks to verify before utilizing a hand-held radiation meter?

1/M Lab Questions:

Answer **TRUE** or **FALSE** to the following:

- a. We increased the number of fission neutron (N_f) in the reactor by lowering control rods into the core.

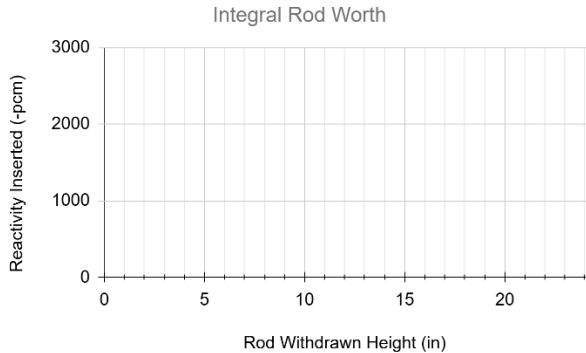
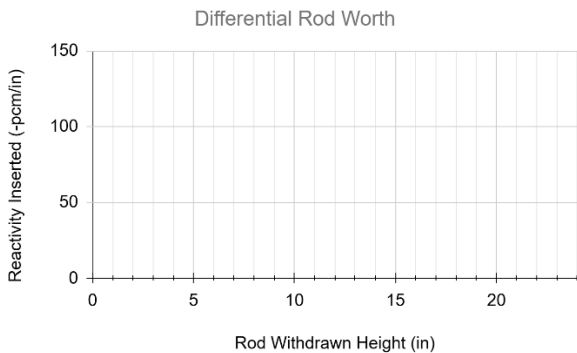
- b. For each rod pull, we pulled halfway from where the pull started to the projected critical rod height **OR** until NFM counts doubled, whichever happened first.

- c. As $CR_i \rightarrow \infty$, $M \rightarrow \infty$, $1/M \rightarrow 0$, and $K_{eff} \rightarrow 1$.

- d. As we got closer to criticality, it took longer for the neutron population to reach subcritical equilibrium.

Control Rod Calibration Lab Questions

1. Draw a sketch of what the differential and integral worth may look like for a Control Rod at the PULSTAR reactor.:



2. Which control rod did we calibrate during the lab?
 - ☐ Safety #1
 - ☐ Safety #2
 - ☐ Regulating Rod
 - ☐ Shim Rod

Equations:

$$\rho = (K_{\text{eff}} - 1) / K_{\text{eff}} \quad \text{where } \rho \text{ is in units of } \Delta k/k.$$

$$K_{\text{eff}} = 1/(1-\rho) \quad \text{where } \rho \text{ is in units of } \Delta k/k.$$

$$\beta_{\text{eff}} = 730 \text{ pcm} = 0.0073 \Delta k/k \text{ (assumed value for PULSTAR)}$$

$$CR_2/CR_1 = (1-K_{\text{eff}1})/(1-K_{\text{eff}2})$$

$$\bar{l} = l_d(\beta) + l_p(1 - \beta)$$

$$P_f = P_o e^{\left(\frac{K_{\text{eff}} - 1}{\bar{l}}\right)t}, \text{ for small reactivity insertions}$$

$$1 \text{ Ci} = 3.7 \times 10^{10} \text{ dis/sec}$$

$$1 \text{ kg} = 2.21 \text{ lbm}$$

$$1 \text{ MW} = 3.41 \times 10^6 \text{ BTU/hr}$$

$$N_A = 6.022 \times 10^{23} \text{ atoms/mol}$$

$$DR = \frac{6 \cdot C \cdot E \cdot n}{d^2}, \text{ where:}$$

DR: Dose rate in R/hr

C: Activity in Curies

E: Energy in MeV

n: Probability of occurrence per decay

d: Distance in ft